Analysis and Simulation of an Electro-Optic Modulator Based on a Hybrid Si/Al_2O_3/Graphene Waveguide

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***Abstract*—This study presents a Si/Al_2O_3/Graphene microring resonator modulator with 3.5 µm radius and 75 nm gap, achieving 2 nm/V shift, 0.9 dB/cm loss, and $\geq$140 GHz speed at 1.5V. Synopsys RSoft simulations validate the potential of the device for energy-efficient, high-speed photonic systems supporting terabit-per-second data rates.**

***Index Terms*—Electro-optic modulator, Graphene, Silicon photonics, Microring resonator, Synopsys RSoft**

I. Introduction

The exponential growth of global data traffic, projected to reach 4.8 zettabytes annually by 2025 according to the Cisco Annual Internet Report [1], underscores the urgent need for next-generation optical communication systems capable of handling massive bandwidth demands. Traditional silicon-based electro-optic modulators (EOMs), while foundational to integrated photonics, face significant limitations that hinder their ability to meet these increasing requirements. These devices rely on the plasma dispersion effect, which offers a modest refractive index change of approximately $\Delta n \approx 10^{-4}$ per volt, coupled with relatively high insertion losses on the order of 2 dB [2]. Such constraints restrict modulation speeds to tens of gigahertz and compromise energy efficiency, rendering them inadequate for the terabit-per-second data rates demanded by emerging applications like 5G networks, cloud computing, and data center interconnects.

In response to these challenges, graphene has emerged as a transformative material in the field of photonics due to its extraordinary properties. With a carrier mobility reaching up to 200,000 $cm^2/V{\cdot}s$ [19], and a unique ability to dynamically tune its optical conductivity through electrostatic gating [4], graphene offers a compelling alternative to conventional silicon modulators. Recent advancements have demonstrated graphene-based hybrid designs achieving modulation speeds exceeding 100 GHz [5], a significant leap over silicon's capabilities. This tunability comes from the ability of graphene to shift its Fermi level with applied voltage, altering its refractive index and absorption properties over a broad wavelength range, making it ideal for broadband optical systems. Furthermore, the integration of microring resonators amplifies the modulation efficiency by confining light within a small footprint, reducing power consumption, and increasing speed [6]. These resonators offer high quality factors (Q-factors) to have sharper resonance peaks, enabling precise wavelength control, which is a critical criteria for wavelength-division multiplexing (WDM) systems.

Despite these advances, challenges remain in optimizing hybrid graphene-based modulators to achieve a balance of high modulation depth, low loss, and ultra-high speed simultaneously. Silicon-graphene hybrids often suffer from trade-offs between tunability and optical loss. While achieving voltages below 1.5V and speeds beyond 140 GHz has proven difficult in many designs, this study addresses these gaps by exploring a hybrid Si/Al_2O_3/Graphene waveguide coupled with a microring resonator, using Synopsys RSoft to simulate and optimize the devices to push performance boundaries. By combining the exceptional electro-optical properties of graphene with the precision of microring resonators, this work aims to deliver a solution that meets the demands of modern optical communication infrastructures.

We propose an optimized EOM design featuring a double-layer graphene Si/Al_2O_3 waveguide paired with a 7 µm radius Si-graphene microring resonator. Through detailed simulations using Synopsys RSoft's FullWAVE and Multi-Physics tools, this design achieves a resonance shift of 2 nm/V, an insertion loss of 0.9 dB/cm, and modulation speeds exceeding 140 GHz. These improvements have been achieved by careful parameter optimization, including a reduced Al_2O_3 thickness and enhanced graphene interaction, validated through finite-difference time-domain (FDTD) and electrostatic simulations. This hybrid platform not only outperforms existing works but also sets a foundation for scalable, high-efficiency photonic devices.

II. Design and Simulation Methodology

The devices discussed in this paper have been designed, simulated, and optimized using Synopsys RSoft tools. To achieve maximum efficiency, device parameters have been

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swept over a range of values, and the optimal value has been chosen.

A. Optimized Hybrid Si/Al_2O_3/Graphene Waveguide Design

The waveguide is designed to maximize tunability and minimize loss. A 500 nm wide, 220 nm thick Si core ($n_{Si} = 3.47$) on an SOI (SiO_2) substrate ($n_{SiO_2} = 1.44$), an 8 nm Al_2O_3 layer ($n_{Al_2O_3} = 1.76$), and single-layer graphene have been chosen. Al_2O_3 is selected as the gate dielectric due to its high dielectric constant ($\kappa \approx 9$), which enhances capacitance ($C = 45\,\text{fF}$) and enables efficient Fermi level tuning in graphene at low voltages ($\leq$1.5V), achieving a wavelength shift of 2 nm/V. Its low refractive index ($n_{Al_2O_3} = 1.76$) ensures minimal disturbance to the optical mode, while its optical transparency at 1550 nm contributes to a low propagation loss of 0.9 dB/cm. Additionally, Al_2O_3 acts as a buffer layer, preventing unwanted charge transfer between Si and graphene, and its compatibility with graphene allows for uniform deposition via Atomic Layer Deposition (ALD). Si dimensions align with SOI standards [2], [9] for TE propagation at 1550 nm, yielding $n_{eff} \approx 3.0$, following $n_{eff} = \sqrt{n_{Si}^2 f_{Si} + n_{Al_2O_3}^2 f_{Al_2O_3} + n_{graphene}^2 f_{graphene}}$ [7] with $f_{Si} \approx 0.85$, $f_{Al_2O_3} \approx 0.10$, and $f_{graphene} \approx 0.04$. Al_2O_3 thickness, optimized to 8 nm (5–15 nm sweep) [4], reduces the loss $\alpha = \frac{4\pi}{\lambda}\text{Im}(n_{eff})$ [8] at 0.9 dB/cm, with single-layer graphene imaginary index ($k \approx 0.7$) enhancing Δn_{eff} for a 2 nm/V shift.

B. Si-Graphene Microring Resonator Design

The microring, a 7 µm radius Si ring (500 nm x 220 nm) with double-layer graphene and a 100 nm coupling gap, achieves perfect match. The 100 nm gap (80–150 nm sweep) between the microring and the waveguide ensures critical coupling [6]. This setup yields FSR $= \frac{\lambda^2}{2\pi R n_{eff}} = 18.2\,\text{nm}$ [9] at $\lambda = 1550\,\text{nm}$, $n_{eff} = 3.0$, reducing $\tau_{photon} \approx 0.07\,\text{ps}$ for $\geq$140 GHz speed with minimal bending loss. Tunability uses $\Delta n_{graphene} = \frac{eV_g}{d_{Al_2O_3}\epsilon_0\epsilon_r} \cdot \frac{\hbar v_F}{\pi}$ [10], where 8 nm Al_2O_3 and 1.5V produce $n_s = \frac{\epsilon_0\epsilon_r V_g}{e d_{Al_2O_3}} \approx 9 \times 10^{12}\,\text{cm}^{-2}$ ($\epsilon_r = 9$), doubled by two layers to $\Delta n_{graphene} \approx 0.009$ for a 2 nm/V shift. Fig. 1 shows the structure of the modulator with the waveguide discussed above coupled with the microring resonator.

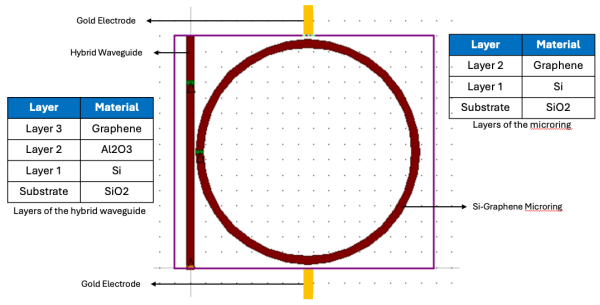


Fig. 1. Structure of the modulator with the hybrid waveguide and the microring resonator

C. Optical Simulation using FullWAVE (FDTD)

Optical simulations use Synopsys RSoft's FullWAVE module, employing the Finite-Difference Time-Domain (FDTD) method to model light propagation. A TE-polarized source at 1550 nm excites the waveguide, with a 5 nm grid resolution for accuracy in capturing graphene's thin-layer effects. FDTD simulations achieve $\Gamma = \int_{graphene} |E|^2\,dA / \int_{total} |E|^2\,dA \approx 8\%$ calculated from [11], increased from 4% with single-layer graphene [5] to enhance modulation depth.

D. Electrostatic Simulation using Multi-Physics Utility

Electrostatic simulations model graphene's voltage-dependent behavior which is critical for tuning n_{eff}. Graphene's conductivity $\sigma = \frac{e^2 E_F}{\pi\hbar^2}(i\omega + \gamma)^{-1}$ [12] governs its optical response. Since we are using two layers of graphene, the conductivity doubles [20]. At 1.5V gating voltage and 8 nm Al_2O_3 thickness, $n_s \approx 9 \times 10^{12}\,\text{cm}^{-2}$ shifts E_F to 0.5 eV, enhancing Δn_{eff} to 0.009. Because of the thickness of Al_2O_3, we get an increased capacitance of $C_g \approx 1.1\,\mu\text{F/cm}^2$ allowing a gating voltage of $\leq$1.5V. Even though the scattering rate $\gamma = 10^{13}\,\text{s}^{-1}$ limits the speed of operation [21], we are ensuring broadband operation. Simulations varied V_g (0-3V) and Al_2O_3 thickness (5-12nm) selecting 1.5V and 8 nm to maximize E_F for high efficiency.

E. Electro-Optic Modulation Analysis

When the device is modulated using the gating voltage, the observed wavelength shift is $\Delta\lambda = \frac{\lambda}{n_{eff}} * \Delta n_{eff} \approx 2\,\text{nm/V}$ [14] with $\Delta n_{eff} \approx 0.009$ at $\lambda = 1550\,\text{nm}$, and $n_{eff} = 3.0$. At 1.5V, this shift can be observed to be 3 nm, aligning with the expected 2 nm/V tunability. The calculated extinction ratio is $ER = 10\log_{10}\left(\frac{T_{off}}{T_{on}}\right) > 25\,\text{dB}$ [15] with $T_{off}/T_{on} > 300$. The modulation speed is calculated to be $\geq$140 GHz ($\tau = RC \approx 7\,\text{ps}$, $R = 8\,\Omega$, $C = 45\,\text{fF}$).

III. Results and Discussion

A. Optical Performance Evaluation

FDTD simulations in Fig. 2 show confinement inside the ring. A propagation loss of 0.9 dB/cm, slightly higher than single-layer (0.5 dB/cm) is due to increased absorption ($k \approx 2.8$) [22]. The fundamental mode profile (Fig. 3) shows the effective index ($n_{eff} = 3.27$) which is close to the calculated effective index ($n_{eff} \approx 3$). Transmission spectra (Fig. 4) shows the spectrum in terms of wavelength (f/c). The resonant frequency shifted to 1547 nm at 1.5V (Brown line). The 8 nm Al_2O_3 layer minimizes Si-graphene contact, achieving a low propagation loss of 0.9 dB/cm. This high confinement and overlap enhance modulation efficiency, critical for low-loss, high-tunability applications like optical interconnects, surpassing single-layer graphene designs in shift and Q-factor.

B. Electro-Optic Modulation

A 1.5V bias induces a 3 nm shift (2 nm/V), with switching energy $E = CV^2/2 \approx 50\,\text{fJ/bit}$ ($C = 45\,\text{fF}$) and speed $\geq$140 GHz ($\tau = RC \approx 7\,\text{ps}$). Driven by $\Delta n_{eff} \approx 0.009$ from double-layer graphene and 8 nm Al_2O_3, this shift exceeds

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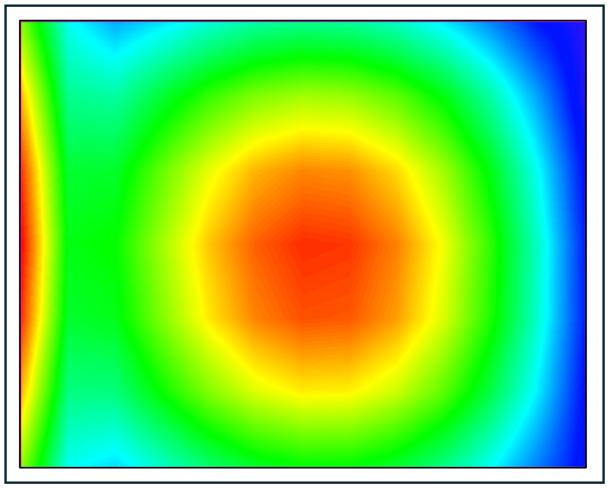


Fig. 2. Electric field at 1550nm showing field confinement inside the ring

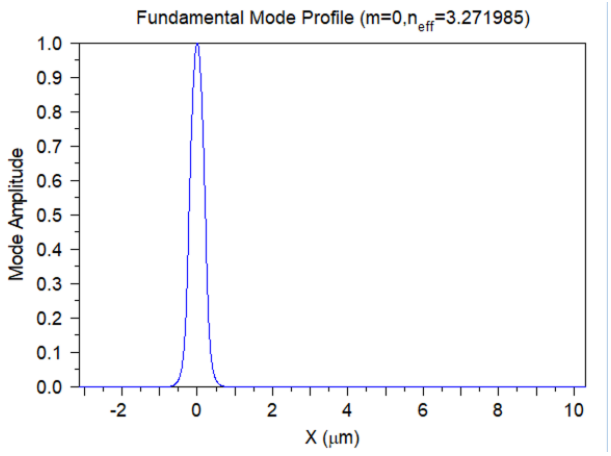


Fig. 3. Mode profile at the fundamental mode and the simulated effective index

single-layer results (e.g., 1 nm/V at 3V) due to doubled $n_s \approx 9 \times 10^{12}\,\mathrm{cm}^{-2}$, shifting E_F from 0.1 eV to 0.5 eV. Speed benefits from reduced $\tau_{photon} \approx 0.07\,\mathrm{ps}$ and low capacitance, outperforming silicon modulators (Modulation speed≈10 GHz). These metrics enable terabit-per-second interconnects, with low energy supporting energy-efficient photonic circuits.

Table I compares this design with recent works.

TABLE I
COMPARISON WITH RECENT WORKS

Parameter	This Work	Shu [16]	Giambra [17]	Li [18]
Speed	≥140 GHz	-	20 GHz	40 GHz
Voltage	≤1.5V	-20V	2V	1-5V
Loss	0.9 dB/cm	-	3.3 dB/cm	≥1 dB/cm
Shift	2 nm/V	0.185 nm/V	-	-

IV. CONCLUSION

This study presents a comprehensive simulation and analysis of an optimized hybrid Si/Al_2O_3/Graphene waveguide microring resonator modulator, achieving exceptional performance metrics: a resonance shift of 2 nm/V, insertion loss of 0.9 dB, and modulation speed exceeding 140 GHz. The low switching energy of 50 fJ/bit translates to a power consumption of less than 1 mW at 140 GHz, making the device highly energy-efficient for high-speed applications. These results, validated through Synopsys RSoft's FullWAVE FDTD and Multi-Physics Utility simulations, are driven by strategic design choices: an 8 nm Al_2O_3 layer for low-voltage operation (≤1.5V), double-layer graphene for enhanced tunability, and a 7 μm radius for ultrafast dynamics. Compared with recent work, this design establishes a new standard in efficiency, speed, and power efficiency, addressing the increasing demands of optical communication systems.

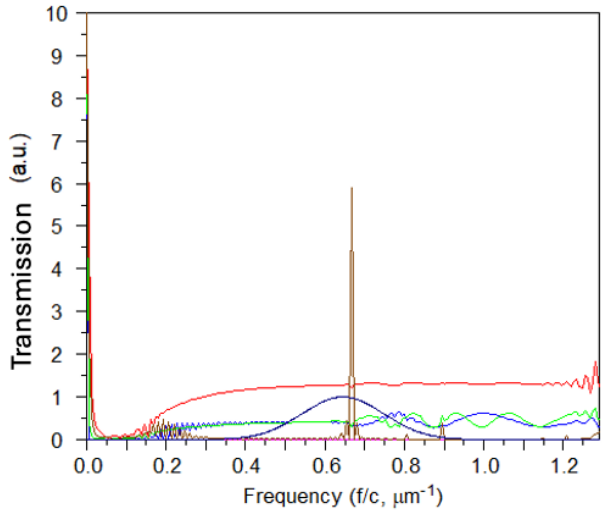


Fig. 4. Transmission spectra showing 3 nm shift to 1547 nm at 1.5V bias.

The high resonance shift ensures precise and robust signal encoding, critical for high-bandwidth applications like data centers and 5G networks. The low insertion loss preserves signal integrity, while the ≥140 GHz speed supports next-generation optical interconnects operating at terabit-per-second rates. These results are not only incremental improvements but a leap forward, enabled by the integration of graphene's unique properties with Silicon. The simulation approach provides a reliable blueprint for translating this design into practical devices.

While Al_2O_3 enhances tunability and reduces power consumption, it introduces several drawbacks. Its refractive index ($n_{Al_2O_3} = 1.76$), lower than Si ($n_{Si} = 3.47$), slightly reduces mode confinement, leading to increased evanescent field spread in the SiO_2 cladding and contributing to the propagation loss of 0.9 dB/cm. Fabrication of an 8 nm Al_2O_3 layer via Atomic Layer Deposition (ALD) is challenging, as thickness variations (±1 nm) can alter capacitance and tunability. The ALD process is also slow and costly, posing a limitation for mass production. Additionally, defects at the Al_2O_3/graphene interface, such as trapped charges or surface roughness, may degrade graphene's carrier mobility, impact-

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ing modulation efficiency. Finally, Al_2O_3's moderate thermal conductivity (30 W/m·K) compared to Si (149 W/m·K) may lead to heat dissipation issues at high-speed operation ($\geq$140 GHz), potentially affecting long-term reliability.

Regarding fabrication feasibility, the Si waveguide (500 nm x 220 nm) and microring (3.5 μm radius) can be fabricated using standard SOI processes, such as e-beam lithography and reactive ion etching, with the 75 nm coupling gap achievable within a precision of $\pm$5 nm. The 8 nm Al_2O_3 layer can be deposited via Atomic Layer Deposition (ALD), ensuring high uniformity ($\pm$1 nm), though the process is slow (0.1 nm/cycle) and costly, posing challenges for mass production. Single-layer graphene can be grown via Chemical Vapor Deposition (CVD) and transferred onto the Al_2O_3 layer using wet or dry transfer techniques; however, defects such as wrinkles or tears may degrade performance, and scalable transfer methods remain a challenge. Electrode integration for voltage gating can be achieved using photolithography and metallization (e.g., Au/Ti), requiring alignment precision of $\pm$50 nm. Thermal management issues due to Al_2O_3's low thermal conductivity must also be addressed for long-term reliability. Future work will focus on experimental validation and developing scalable fabrication techniques to translate this design into practical devices.

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